Consider again

$$A\bar{x} = \bar{b}$$

$$A\tilde{x} = \tilde{b}$$

$$\bar{r} = \bar{b} - A\tilde{x} = \bar{b} - \tilde{b}$$

$$\bar{e} = \bar{x} - \tilde{x}$$

$$A\bar{e} = \bar{r}$$

Thm We have

$$\frac{1}{\kappa(A)} \frac{\|\bar{r}\|}{\|\bar{b}\|} \leq \frac{\|\bar{e}\|}{\|\bar{x}\|} \leq \kappa(A) \frac{\|\bar{r}\|}{\|\bar{b}\|}$$

Pf done above.

Next, $\cancel{\dfrac{\|\bar{x} - \tilde{x}\|}{\|\bar{x}\|}}$ $\|\bar{r}\|\|\bar{x}\| = \|A\bar{e}\|\,\|A^{-1}\bar{b}\| \leq$

$$\leq \|A\|\|A^{-1}\|\,\|\bar{e}\|\|\bar{b}\| = \kappa(A)\|\bar{e}\|\|\bar{b}\|$$

$$\Downarrow$$

$$\frac{1}{\kappa(A)} \frac{\|\bar{r}\|}{\|\bar{b}\|} \leq \frac{\|\bar{e}\|}{\|\bar{x}\|}$$

$\square$.

Neumann Series and Iterative Refinement

Convergence of sequence of vectors in a normed space:

$$\bar{v}^{(k)} \underset{k\to\infty}{\rightarrow} \bar{v} \iff \lim_{k\to\infty} \|\bar{v}^{(k)} - \bar{v}\| = 0$$

Facts:

1) all norms in a <u>finite-dimensional</u> vector space are equivalent, that is, for any norms $|\cdot|$, $\|\cdot\|$, there constants C_1, C_2 such that

$$c_1 \|\bar{v}\| \le |\bar{v}| \le C_2 \|\bar{v}\| \quad \forall \bar{v} \in V.$$

2) Every Cauchy sequence has a limit

Def A sequence $\{v^{(i)}\}_{i=1}^{\infty}$ is Cauchy iff.

$$\lim_{k\to\infty} \sup_{i,j\geq k} \|v^{(i)} - v^{(j)}\| = 0$$

or equivalently,

$\forall \varepsilon > 0 \ \exists K \in \mathbb{Z}$, such that

$$\|v^{(i)} - v^{(j)}\| < \varepsilon \quad \forall i,j \geq K$$

Thm if $A_{n\times n}$ and $\|A\| < 1$, then $I - A$ is invertible, and

$$(I-A)^{-1} = \sum_{k=0}^{\infty} A^k = I + A + A^2 + \dots$$

Pf 1) If $I-A$ is not invertible, then $\exists\, \bar{x} : \|\bar{x}\| = 1$ and

$$(I-A)\bar{x} = 0 \iff A\bar{x} = \bar{x}$$

Then

$$1 = \|\bar{x}\| = \|A\bar{x}\| \le \|A\|\,\|\bar{x}\| = \|A\|$$

so $\|A\| \geq 1$, but $\|A\| < 1$ by assumption $\Rightarrow$ contradiction.

$$\Rightarrow \det(I-A) \neq 0.$$

2) show that $\sum_{k=0}^{\infty} A^k$ converges to $(I-A)^{-1}$.

We are going to show that

$$(I-A)\left(\sum_{k=0}^{m} A^k\right) \to I \quad \text{as } m \to \infty$$

$$(I-A)\left(\sum_{k=0}^{m} A^k\right) = \sum_{k=0}^{m} A^k - \sum_{k=0}^{m} A^{k+1} = I - A^{m+1}$$

Prove that $\lim_{m\to\infty} A^{m+1} = 0$.

Indeed. $\| A^{m+1} \| \le \| A \|^{m+1}$

and $\|A\| < 1 \Rightarrow \|A\|^{m+1} \to 0$ as $m \to \infty$ $\Rightarrow \| A^{m+1} \| \to 0$ as $m \to \infty$.

$\square$.

Thm Let $A, B_{(n\times n)}$ be such that $\| I - AB \| < 1$. Then A, B are invertible, and

$$A^{-1} = B \sum_{k=0}^{\infty} (I - AB)^k$$

$$B^{-1} = \left[\sum_{k=0}^{\infty} (I - AB)^k \right] A$$

Pf. By the previous thm, the matrix

$$I - (I - AB) = AB$$

is invertible, and

$$(AB)^{-1} = \sum_{k=0}^{\infty} (I - AB)^k$$

if AB is invertible then A, B are (separately invertible since $\det(AB) = = \det A \cdot \det B$

Then $A^{-1} = B(\cancel{A}B)^{-1} = \underbrace{BB^{-1}}_{I} A^{-1} = B \sum_{k=0}^{\infty} (I - AB)^k$

$$B^{-1} = B^{-1} A^{-1} A = (AB)^{-1} A = \left[\sum_{k=0}^{\infty} (I - AB)^k \right] A.$$

$\square$

Iterative Refinement

Let $\bar{x}^{(0)}$ be an approximate solution of $A\bar{x} = \bar{b}$

Then the exact solution

$$\bar{x} = \bar{x}^{(0)} + A^{-1}\left(\bar{b} - A\bar{x}^{(0)}\right) = \bar{x}^{(0)} + \bar{e}^{(0)}$$

Think that $\bar{x}^{(0)} = B\bar{b}$

where $B \approx A^{-1}$.

Then try to update recursively

$$\bar{x}^{(k+1)} = \bar{x}^{(k)} + B\left(\bar{b} - A\bar{x}^{(k)}\right), \quad k \geq 0$$

$$\bar{x}^{(1)} = \bar{x}^{(0)} + B\left(\bar{b} - A\bar{x}^{(0)}\right)$$

$$\bar{x}^{(2)} = \bar{x}^{(1)} + B\left(\bar{b} - A\bar{x}^{(1)}\right)$$

etc.

Suppose that $(I - AB)$ is small : $\| I - AB \| < 1$.

Then, by the thm, $A^{-1} = B \sum_{k=0}^{\infty} (I - AB)^k$

$$\bar{x}^{(0)} = B\bar{b}$$

$$\bar{x} = \bar{x}^{(0)} + A^{-1}(\bar{b} - A\bar{x}^{(0)}) = B\bar{b} + A^{-1}\bar{b} -$$

$$-AB\bar{b} = (B + A^{-1} - AB)\bar{b} \overset{\text{thm}}{=}$$

$$= \left(B + B\left[\sum_{k=0}^{\infty} (I - AB)^k\right] - AB\right)\bar{b}$$

$$\bar{x} = B \sum_{k=0}^{\infty} (I - AB)^k \bar{b}$$

We would like to show that

$$\bar{x}^{(m)} = B \sum_{k=0}^{m} (I - AB)^k \bar{b} \qquad (*)$$

correspond to $\bar{x}^{(m+1)} = \bar{x}^{(m)} + B(\bar{b} - A\bar{x}^{(m)})$

(and thus $\bar{x}^{(m)} \underset{m \to \infty}{\to} \bar{x}$ if $\| I - AB \| < 1$

Pf. Math. induction :

if $P(a)$ is a property

$a \in \mathbb{Z}$, $a \geq a_0 \in \mathbb{Z}$.

then math. induction works like this.

1. Check directly that $P(a_0)$ is true
2. Prove that $P(a)$ true for general a implies that $P(a+1)$ is true.

$$\Downarrow$$

$P(a)$ is true for $\forall a \geq a_0$, $a \in \mathbb{Z}$.

$P(m)$: $\bar{x}^m = B \sum_{k=0}^{m} (I - AB)^k \bar{b}$

corresponds to $\bar{x}^{(\ell+1)} = \bar{x}^{(\ell)} + B(\bar{b} - A\bar{x}^{(\ell)})$,

$\bar{x}^{(0)} = B\bar{b}$

$\ell = 0, 1, 2 \ldots, m-1$

1. $P(0)$: $\bar{x}^{(0)} = B\bar{b}$ - holds
2. Assume $P(m)$ is true, use iteration to update:

$$\bar{x}^{(m+1)} = \bar{x}^{(m)} + B(\bar{b} - A\bar{x}^{(m)}) =$$

$$= B\left[\sum_{k=0}^{m} (I-AB)^k\right] \bar{b} + B\bar{b} -$$

$$- BAB\left[\sum_{k=0}^{m} (I-AB)^k\right]\bar{b} =$$

$$= B\left\{ \bar{b} + \left[\sum_{k=0}^{m} (I-AB)^k - AB\sum_{k=0}^{m} (I-AB)^k\right] \bar{b} \right\} =$$

$$= B\left\{ \bar{b} + (I-AB)\sum_{k=0}^{m} (I-AB)^k \bar{b} \right\} =$$

$$= B\left\{ \bar{b} + \sum_{k=1}^{m+1} (I-AB)^k \bar{b} \right\} =$$

$$= B\left\{ (I-AB)^0 \bar{b} + \sum_{k=1}^{m+1} (I-AB)^k \bar{b} \right\} =$$

$$= B\sum_{k=0}^{m+1} (I-AB)^k \bar{b} \iff P(m+1)$$

$$= \bar{x}^{(m+1)}$$

$\square$.

Solution by iterative methods.

Idea: $A\bar{x} = \bar{b}$ (1)

Choose a splitting matrix Q

and write

(2) $Q\bar{x} = (Q-A)\bar{x} + \bar{b}$

if Q^{-1} exists

$\bar{x} = \underbrace{Q^{-1}(Q-A)\bar{x} + Q^{-1}\bar{b}}_{T(\bar{x})}$

so $\bar{x}$ is a fixed point of $T(\bar{x})$

Solve (2) by iteration:

(3) $Q\bar{x}^{(k)} = (Q-A)\bar{x}^{(k-1)} + \bar{b}$.

To get $\bar{x}^{(k)}$ from (3), need to "invert" Q, so

$\bar{x}^{(k)} = Q^{-1}(Q-A)\bar{x}^{(k-1)} + Q^{-1}\bar{b} =$

$= T\left(\bar{x}^{(k-1)}\right)$ – converges to the fixed point $\bar{x}$ if T is contractive

Q is chosen so that

1. $\bar{x}^{(k)}$ can be computed easily
   (Q^{-1} can be found cheaply)
2. $\bar{x}^{(k)} \to \bar{x}$ rapidly

1. $\Rightarrow$ $Q\bar{x} = \bar{y}$ must be ~~easy~~ easy to solve
2. $\Rightarrow$ $Q^{-1} \approx A^{-1}$

conflicting

Assume that A^{-1} and Q^{-1} exist.

Write (2)

$$\bar{x}^{(k)} = Q^{-1}(Q - A)\,\bar{x}^{(k-1)} + Q^{-1}\bar{b} =$$

$$= (I - Q^{-1}A)\,\bar{x}^{(k-1)} + Q^{-1}\bar{b}$$

Note that the exact solution (the fixed point, or $\bar{x} = \lim\limits_{k \to \infty} \bar{x}^{(k)}$), must solve (if that is true)

$$\bar{x} = (I - Q^{-1}A)\,\bar{x} + Q^{-1}\bar{b}$$

Then

$$\bar{x} - \bar{x}^{(k)} = \left(I - Q^{-1}A\right)\left(\bar{x} - \bar{x}^{(k-1)}\right)$$

$$\Downarrow$$

$$\|\bar{x} - \bar{x}^{(k)}\| \leq \left\|\left(I - Q^{-1}A\right)\left(\bar{x} - \bar{x}^{(k-1)}\right)\right\| \leq$$

$$\leq \left\|\left(I - Q^{-1}A\right)\right\| \, \|\bar{x} - \bar{x}^{(k-1)}\| \underset{\text{repeatedly}}{\overset{\text{apply}}{\leq}} \dots$$

$$\dots \leq \|I - Q^{-1}A\|^k \, \|\bar{x}^{(0)} - \bar{x}\|$$

if $\|I - Q^{-1}A\| < 1$ $\quad \left(Q^{-1} \approx A^{-1} \text{ or } Q^{-1}A \approx I \right)$

then $\lim\limits_{k\to\infty} \|I - QA\|^k = 0 \Rightarrow$

$$\Rightarrow \|\bar{x} - \bar{x}^{(k)}\| \to 0 \quad \text{as} \quad k \to \infty$$

$$\Leftrightarrow \bar{x}^{(k)} \to \bar{x} \quad \text{as} \quad k \to \infty.$$

Thm If $\| I - Q^{-1} A \| < 1$ ~~for~~ some subordinate matrix norm then the sequence produced by

$$Q \bar{x}^{(k)} = (Q - A)\bar{x}^{(k-1)} + \bar{b}, \quad (k \geq 1)$$

converges to the exact solution of $A\bar{x} = \bar{b}$ for any initial vector $\bar{x}^{(0)}$.

(and in this contraction mapping then yields aposteriori error estimate

$$\| \bar{x}^{(k)} - \bar{x} \| \leq \frac{\delta}{1-\delta} \| \bar{x}^{(k)} - \bar{x}^{(k-1)} \| \quad (4)$$

when $0 < \delta < 1$ is the upper bound for $\| I - Q^{-1} A \|$.

Specific methods

1) Richardson

$$Q = I$$

$$\bar{x}^{(k)} = (I - A)\bar{x}^{(k-1)} + \bar{b}$$

2) Jacobi. Q is diagonal.

with $q_{ii} = a_{ii}$

$$Q = \begin{pmatrix} a_{11} & & & 0 \\ & a_{22} & & \\ & & \ddots & \\ 0 & & & a_{nn} \end{pmatrix}, \quad A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

$$Q\bar{x}^{(k)} = (Q - A)\bar{x}^{(k-1)} + b$$

$$\boxed{i = 1, 2, \ldots, n} \quad x_i^{(k)} = \frac{1}{a_{ii}} \left((Q - A)_{il} x_l^{(k-1)} + b_i \right)$$

3) Gauss - Seidel.

Q is lower triangular part of A including the diagonal

Jacobi method:

$$Q = \begin{bmatrix} a_{11} & & & \\ & a_{22} & 0 & \\ & 0 & \ddots & \\ & & & a_{nn} \end{bmatrix}$$

$$Q\bar{x}^{(k)} = (Q - A)\bar{x}^{(k-1)} + \bar{b}$$

$$x_i^{(k)} = \frac{1}{a_{ii}} (Q-A)_{il} \underbrace{x_l^{(k-1)}}_{} + b_i$$

summation over l.

Note: $Q^{-1}A$ has elements

$$\frac{a_{ij}}{a_{ii}} \qquad i = \overline{1,n},\ j = \overline{1,n}$$

and $Q^{-1}A$ has

1 on the main diagonal.

$\Downarrow$

$$\|I - Q^{-1}A\|_\infty = \max_{1\le i\le n} \sum_{\substack{j=1 \\ j\ne i}}^{n} |a_{ij}/a_{ii}| \qquad (*)$$

Thm If A is diagonally dominant, then the sequence produced by Jacobi iterations converges to the solution of $A\bar{x} = \bar{b}$ for any starting vector

Pf Diag. dominance:

$$|a_{ii}| > \sum_{\substack{j=1\\ j\neq i}}^{n} |a_{ij}|, \quad (1 \le i \le n)$$

$$\Downarrow$$

$$\sum_{\substack{j=1\\ j\neq i}}^{n} \left|\frac{a_{ij}}{a_{ii}}\right| < 1$$

$$\Downarrow (*)$$

$$\|I - Q^{-1}A\|_\infty < 1.$$

□.

$$Q x^{(k+1)} = (Q - A) x^{(k)} + b$$

$$\bar{x}^{(k+1)} = Q^{-1}(Q-A)\,\bar{x}^{(k)} + Q^{-1}\bar{b}$$

$$\bar{x}^{(k+1)} = (I - Q^{-1}A)\,\bar{x}^{(k)} + Q^{-1}\bar{b}$$

$$\Downarrow \quad \|I - Q^{-1}A\| < 1$$

$$\bar{x}^{(k)} \to \bar{x} : \quad \bar{x} = A^{-1}\bar{b}$$

Analysis of iterative methods.

Consider $\bar{x}^{(k)} = G\,\bar{x}^{(k-1)} + \bar{c}$

$G_{n\times n}$ - given, $\bar{c}_{n\times 1}$ - given

(general linear iteration).

Background:

e-values: complex #s, λ such that

$A - \lambda I$ is not invertible

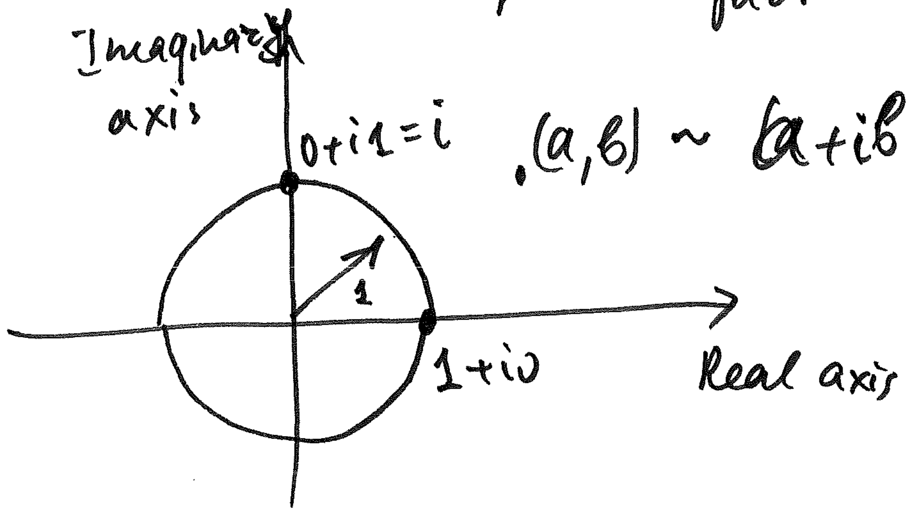
$$\Updownarrow$$

$$\det(A - \lambda I) = 0 \;-$$

Complex #s: $a+ib$, $a, b \in \mathbb{R}$

a ↑ real part, ib ↑ imaginary part

$i^2 = -1.$



$(a,b) \sim a+ib$

unit circle : $\{x+iy : x^2+y^2=1\}$

$x = \cos\alpha$
$y = \sin\alpha$ for some α

<u>Spectral radius</u>

$$\rho(A) = \max\{|\lambda| : \det(A-\lambda I) = 0\}$$

<u>Def</u> $|x+iy| = \sqrt{x^2+y^2}$

Similarity. A is similar to B if there is a non-singular S such that

$$S^{-1} A S = B$$

(!!) Similar matrices have the same e-values.

Thm Every square matrix is similar to an (possibly complex) upper triangular matrix whose off-diagonal entries are arbitrary small

Pf. By Schur's thm, any

$A_{n\times n}$ is similar to $T_{n\times n}$

(upper-triangular) (T may be complex)

Note that upper triangular (lower) matrix has its e-values on the main diagonal

Let $0<\varepsilon<1$ and let $D =$

$= \operatorname{diag}(\varepsilon, \varepsilon^2, \varepsilon^3, \dots \varepsilon^n)$

Consider $V = D^{-1} T D$. The generic elements of V are $t_{ij}\,\varepsilon^{j-i}$ and V is upper triangular.

$$\underset{D^{-1}}{\begin{pmatrix} \varepsilon^{-1} & & & 0 \\ & \varepsilon^{-2} & & \\ & & \varepsilon^{-3} & \\ 0 & & & \ddots \\ & & & & \varepsilon^{-n} \end{pmatrix}} \underset{T}{\begin{pmatrix} t_{11} & & * \\ & t_{22} & \\ 0 & & \ddots \\ & & & t_{nn} \end{pmatrix}} \underset{D}{\begin{pmatrix} \varepsilon & & & 0 \\ & \varepsilon^{2} & & \\ 0 & & \varepsilon^{3} & \\ & & & \ddots \\ & & & & \varepsilon^{n} \end{pmatrix}}$$

$$(TD)_{kl} = \sum_{m=1}^{n} T_{km} D_{ml} = \sum_{m \geq k}^{n} t_{km} D_{ml} =$$

$$= \sum_{m \geq k}^{n} t_{km} \delta_{ml} \varepsilon^{l} = \varepsilon^{l} \sum_{m \geq k}^{n} t_{km} \delta_{ml} =$$

$$\delta_{ml} = \begin{cases} 1 & \text{if } m = l \\ 0 & \text{otherwise} \end{cases}$$

$k = \overline{1,n}$

$l = \overline{1,n}$

$$= \varepsilon^{l} \begin{cases} t_{kl} & \text{if } l \geq k \\ 0 & \text{otherwise} \end{cases}$$

$$(TD)_{kl} = \begin{cases} \varepsilon^{l} t_{kl} & \text{if } l \geq k \\ 0 & \text{if } \cancel{\text{not}}\ l < k \end{cases}$$

$$TD = \begin{pmatrix} t_{11}\varepsilon & t_{12}\varepsilon^{2} & \dots & t_{1n}\varepsilon^{n} \\ & T_{22}\varepsilon^{2} & \dots & T_{2n}\varepsilon^{n} \\ & & \ddots & \vdots \\ 0 & & & T_{nn}\varepsilon^{n} \end{pmatrix}$$

$$D^{-1}TD = \begin{pmatrix} \varepsilon^{-1} & & & \\ & \varepsilon^{-2} & & 0 \\ 0 & & \ddots & \\ & & & \varepsilon^{-n} \end{pmatrix} \begin{pmatrix} T_{11}\varepsilon & T_{12}\varepsilon^{2} & \cdots & T_{1n}\varepsilon^{n} \\ & T_{22}\varepsilon^{2} & \cdots & T_{2n}\varepsilon^{n} \\ 0 & & \ddots & \vdots \\ & & & T_{nn}\varepsilon^{n} \end{pmatrix}$$

$$= \begin{pmatrix} T_{11} & T_{12}\varepsilon^{2}\cdot\varepsilon^{-1} & T_{13}\varepsilon^{3}\cdot\varepsilon^{-1} & \cdots & T_{1n}\varepsilon^{n}\cdot\varepsilon^{-1} \\ & T_{22} & T_{23}\varepsilon^{3}\cdot\varepsilon^{-2} & \cdots & T_{2n}\varepsilon^{n}\cdot\varepsilon^{-2} \\ 0 & & \ddots & T_{n-1,n-1} & T_{n-1,n}\varepsilon^{n}\cdot\varepsilon^{-(n-1)} \\ & & & & T_{nn} \end{pmatrix}$$

$$\Downarrow$$

$$(D^{-1}TD)_{ij} = t_{ij}\,\varepsilon^{j-i}$$

$$\Downarrow$$

$$\left| t_{ij}\,\varepsilon^{j-i} \right| \leq \varepsilon \left| t_{ij} \right| \quad \text{when } i \neq j$$

$\varepsilon < 1$

can be made arbitrarily small.

$\square$

Thm $\rho(A)$ satisfies

$$\rho(A) = \inf_{\|\cdot\|} \|A\|$$

where infimum is taken over all subordinate norms.

Fact: $\rho(A) = \lim_{n\to\infty} \|A^n\|^{\frac{1}{n}}$

Pf i. Show that

$$\boxed{\rho(A) \le \inf_{\|\cdot\|} \|A\|}$$

Let λ be an e-value of A. Then

$$\lambda \bar{x} = A\bar{x} \quad \text{for some } x$$

$$\Downarrow$$

$$|\lambda| \|\bar{x}\| = \|A\bar{x}\| \le \|A\| \|\bar{x}\|$$

$$\Downarrow$$

$$|\lambda| \le \|A\|$$

$$\Downarrow$$

$$\rho(A) = \max |\lambda| \le \|A\|$$

Take infimum over all norms

$$\Downarrow$$

$$\rho(A) \leq \inf_{\|\cdot\|} \|A\|.$$

2. Show that $\boxed{\rho(A) \geq \inf_{\|\cdot\|} \|A\|}$

$\forall\, \varepsilon > 0$ there is a matrix S (non-singular) such that

$$S^{-1} A S^{*} = D + T$$

where D is diagonal, T is strictly upper triangular and $\|T\|_\infty \leq \varepsilon$

$$\Downarrow$$

triangle inequality

$$\|S^{-1} A S\|_\infty = \|D + T\|_\infty \leq \|D\|_\infty +$$

$$+ \|T\|_\infty$$

Since D has e-values of A as its entries, $\|D_\infty\| = \max_{1 \leq i \leq n} |\lambda_i| = \rho(A)$

$$\Downarrow$$

$$\|S^{-1}AS\|_\infty \le \rho(A)+\varepsilon$$

Note that $\|S^{-1}AS\|_\infty$ is a subordinate matrix norm of A. Denote this norm by $\|A\|'_\infty$

$$\Downarrow$$

$$\|A\|'_\infty \le \rho(A)+\varepsilon$$

$\Downarrow$ take inf. over all norms on the left

$$\inf_{\|\cdot\|} \|A\| \le \rho(A)+\varepsilon \quad \text{where } \varepsilon>0 \text{ is arbitrary}$$

sending $\varepsilon \to 0$ we obtain

$$\inf_{\|\cdot\|} \|A\| \le \rho(A)$$

Apply the above to analysis of convergence of iterative method.

Thm In order that the interations obtained by

$$\bar{x}^{(k)} = G\bar{x}^{(k-1)} + \bar{c} \qquad (**)$$

produce a sequence converging to $(I-G)^{-1}\bar{c}$ for any starting vector $\bar{x}^{(0)}$, it is <u>necessary</u> and <u>sufficient</u> that $\rho(G) < 1$.

---

previous result: if $\|G\| < 1$ then $(**)$ converges to $(I-G)^{-1}\bar{c}$

(!) The system we are solving is

$$(I-G)\bar{x} = \bar{c} \Rightarrow \text{if } A\bar{x} = \bar{c} \text{ then } G = I - A$$

Pf 1) Suppose $\rho(G) < 1$

By the previous thm, there is a subordinate norm such that

$$\|G\| < 1$$

$$\bar{x}^{(1)} = G\bar{x}^{(0)} + \bar{c}$$

$$\bar{x}^{(2)} = G\bar{x}^{(1)} + \bar{c} = G(G\bar{x}^{(0)} + \bar{c}) + \bar{c} =$$

$$= G^2\bar{x}^{(0)} + G\bar{c} + \bar{c}$$

$$\bar{x}^{(3)} = G\bar{x}^{(2)} + \bar{c} = G(G^2\bar{x}^{(0)} + G\bar{c} + \bar{c}) + \bar{c} =$$

$$= G^3\bar{x}^{(0)} + G^2\bar{c} + G\bar{c} + \bar{c}$$

. . . .

$$\bar{x}^{(k)} = G^k\bar{x}^{(0)} + \sum_{j=0}^{k-1} G^j\bar{c}$$

$$\|G^k\bar{x}^{(0)}\| \le \|G^k\|\,\|\bar{x}^{(0)}\| \le \|G\|^k\,\|\bar{x}^{(0)}\|$$

$\to 0$ as $k \to \infty$ since $\|G\| < 1$

$$\sum_{j=0}^{k-1} G^j \bar{c} \xrightarrow[k\to\infty]{} (I-G)^{-1}\bar{c} = \sum_{j=0}^{\infty} G^j \bar{c}$$

$$\Downarrow$$

$$\bar{x}^{(k)} \to (I-G)^{-1}\bar{c}$$

2. Suppose that the iterations converge to $(I-G)^{-1}\bar{c}$, but $\rho(G) \geq 1$.

Select $\bar{u}$ and λ so that

$$G\bar{u} = \lambda \bar{u}, \quad |\lambda| \geq 1, \quad \bar{u} \neq 0$$

let $\bar{c} = \bar{u}$ and $x^{(0)} = 0$.

Then $\bar{x}^{(k)} = \sum_{j=0}^{k-1} G^j \bar{u} = \sum_{j=0}^{k-1} \lambda^j \bar{u}$

If $\lambda = 1$, then $\bar{x}^{(k)} = k\bar{u} \to$ diverges as $k \to \infty$

If $\lambda \neq 1$, then $\cancel{\bar{x}^{(k)} \to \bar{u} \cdot \sum_{j=0}^{\infty} \lambda^j}$

$$\bar{x}^{(k)} = \bar{u} \sum_{j=0}^{k-1} \lambda^j = \bar{u} \frac{\lambda^k - 1}{\lambda - 1}$$

if $\lambda \neq 1$, $|\lambda| \geq 1$, then $\frac{\lambda^k - 1}{\lambda - 1}$ diverges

since λ^k diverges as $k \to \infty$

$\Downarrow$ ▨.

Contradiction with assumed

convergence of $\bar{x}^{(k)}$ as $k \to \infty$.

$\Rightarrow \rho(A) < 1$.

$\square$.